

Project SENTINEL: Mechanistically Interpretable Steering of Foundation Models for De Novo Protein Binder Design

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Engineered protein biosensors hold transformative potential for real-time disease detection, but their development is limited by dependence on generative protein design models that function as black boxes—candidate binders are produced, yet the internal reasoning distinguishing a high-affinity binder from a poor one is scarcely characterized. This opacity limits rational design improvement, failure-mode diagnosis, and adaptation across disease contexts. Project SENTINEL addresses this by introducing a mechanistically interpretable framework for de novo protein binder design. We train sparse autoencoders (SAEs) on ESM-C, a state-of-the-art protein language model, to extract human-readable features from residue-level activations of Boltz-designed binder candidates. As proof-of-concept, we validated this pipeline on Vilip-1, a blood-circulating biomarker of acute neurological injury detectable at low nanomolar concentrations before greater damage occurs. Diversifying training data—incorporating real, evolutionarily distinct sequences and binder designs against additional targets (UCH-L1, B-FABP)—substantially improved generalization beyond synthetic designs alone. Independently trained SAE dictionaries, including one incorporating binder-target co-attention context via ESM-C’s native chain-break token, converge on the same residues in real proteins, corroborated by InterPro domain annotations and calibrated LLM-assisted labeling—evidence of genuine structural signal, not training artifacts. This interpretability foundation is the necessary precursor to our ultimate goal: steering generative design toward high-affinity binding features—analogueous to concept-guided steering in text-to-image diffusion models—rather than blind latent-space sampling. We are extending this pipeline to UCH-L1, S100B, and B-FABP, SENTINEL’s three clinical targets, toward establishing mechanistic interpretability as a universal accelerant for biosensor design across any disease where early detection saves lives.